LOSSLESS & LOSSY CMPRSN

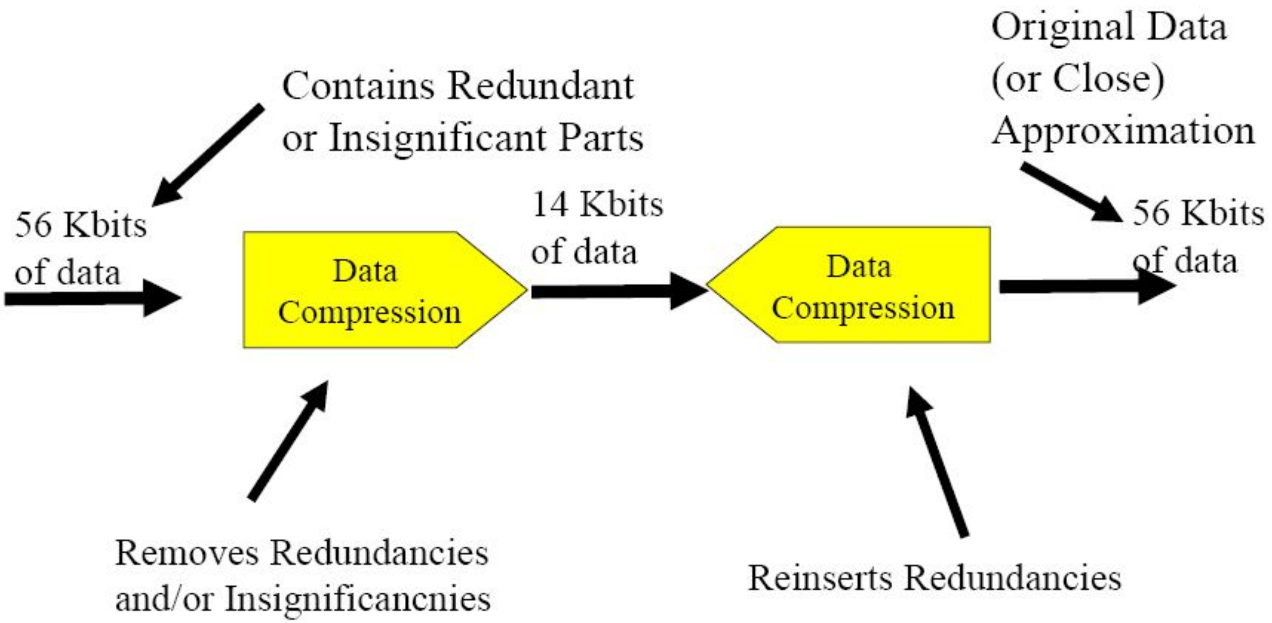
M.Sc. Julio Santisteban Pablo

Lossless & LOSSy Cmprsn

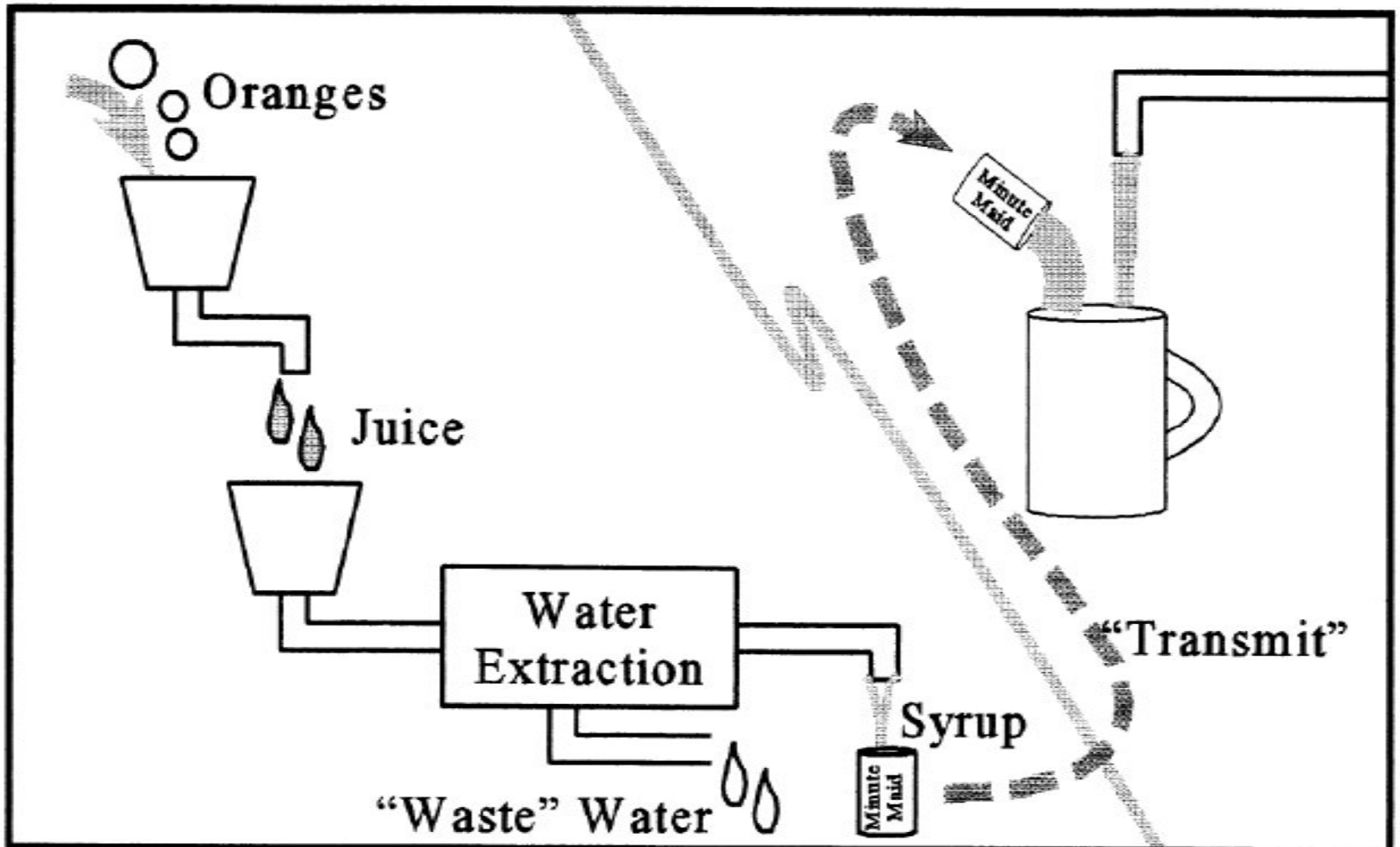
- An Application of Binary Trees and Priority Queues

Introduction

- **Wht is Dta Cmprsn?** The systematic removal and, later, reintroduction of redundancy from data, and possibly also the removal of insignificant data.



Similar to Orange Juice Concentrate:



Lossless Vs. Lossy

- There are two broad categories of data compression:
  - - Lossless Compression
    - Huffman
    - LZW (Lempel-Ziv-Welch)
  - - Lossy Compression
    - DCT (Discrete Cosine Transform)
    - Wavelet

Lossless Compression

- Lossless Compression: Original digital data can be exactly reconstructed from the compresssed data
- Generally applied to:
  - text files (documents, source code, etc.)
  - program files
- Can achieve modest compression ratios:
  - 2:1 to 8:1 depending on type of content of file
- Examples:
  - UNIX “compress” command
  - PKZIP
  - Ram Doublers, Stacker, etc.

Lossy Compression

- Lossy Compression: Original digital data can only be
- approximately reconstructed from the compressed data
- Generally applied to:
  - Signals (speech, music, sensor signals, etc.)
  - Images (digitized photos, digitized video)
- Can achieve large compression ratios:
  - up to (many tens): 1 depending on type & content to data
- Examples:
  - JPEG (for digitized photos)
  - MPEG (for digitized video)
  - Speech compression used in digital cell phones
  - Digital answering machines
  - MP3 (for music)

- Huffman

- Homework
  - LZW (Lempel-Ziv-Welch)
  - Pseudocode

Discrete Cosine Transform

- A discrete cosine transform (DCT) expresses a sequence of finitely many data points in terms of a sum of cosine functions oscillating at different frequencies. DCTs are important to numerous applications in science and engineering, from lossy compression of audio (e.g. MP3) and images (e.g. JPEG)

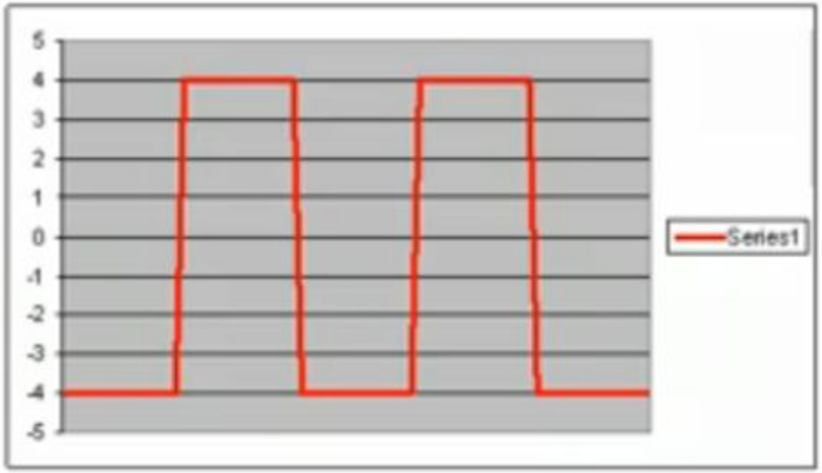
Discrete Cosine Transform

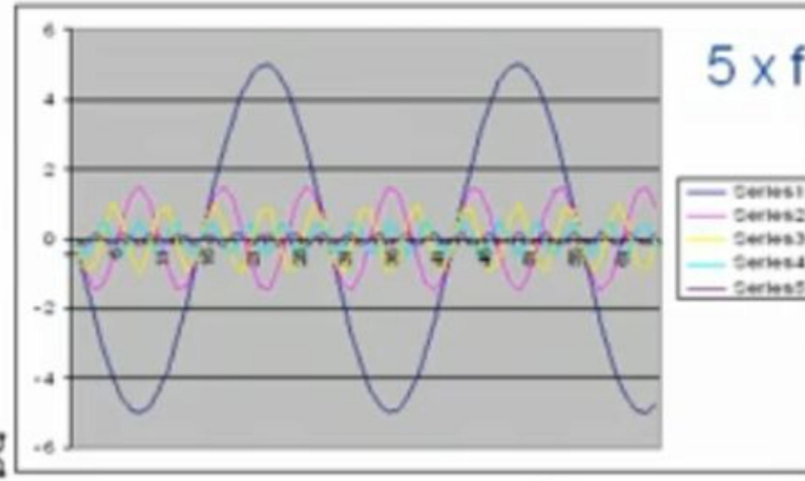
- A discretised version of the Fourier transform
  - Suited to representing spatially quantised (ie raster) images in a frequency quantised (ie tabular) format.
  - Mathematically, the DCT of a function f ranging over a discrete variable x (omitting various important constants) is given by
    - $F(n) = \Sigma x\ f(x) \cos(n\pi x)$
  - Of course, we're usually interested in two-dimensional images, and hence need the two-dimensional DCT, given (omitting even more important constants) by
    - $F(m,n) = \Sigma x\ \Sigma y f(x,y) \cos(m\pi x) \cos(n\pi y)$

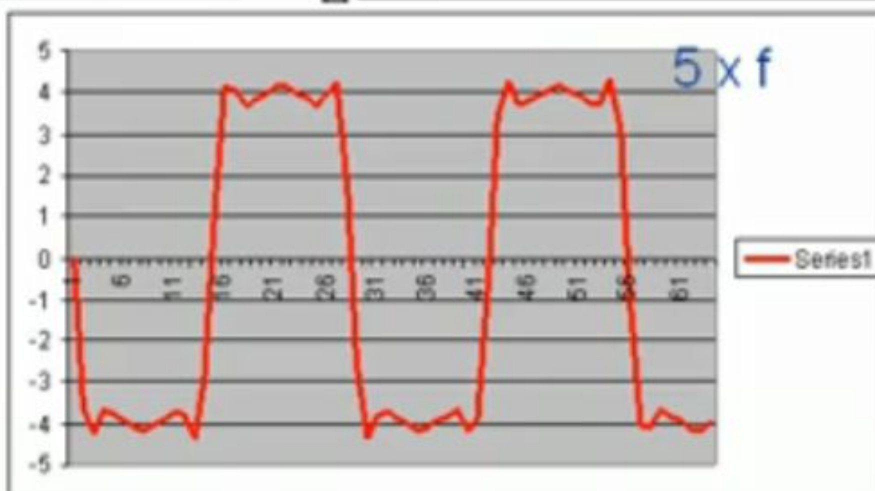
DCT Formula

$$B(k_1, k_2) = \sum_{i=0}^{N_1-1} \sum_{j=0}^{N_2-1} 4 \cdot A(i,j) \cdot \cos\left[\frac{\pi \cdot k_1}{2 \cdot N_1} \cdot (2 \cdot i + 1)\right] \cdot \cos\left[\frac{\pi \cdot k_2}{2 \cdot N_2} \cdot (2 \cdot j + 1)\right]$$

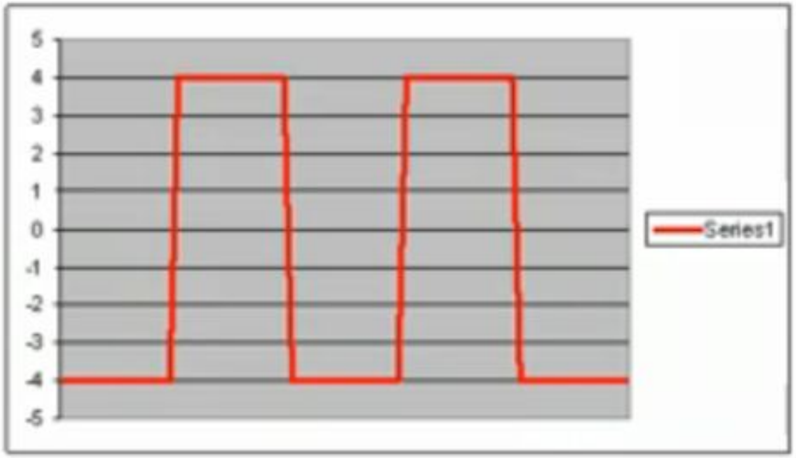
Graphical representation DCT

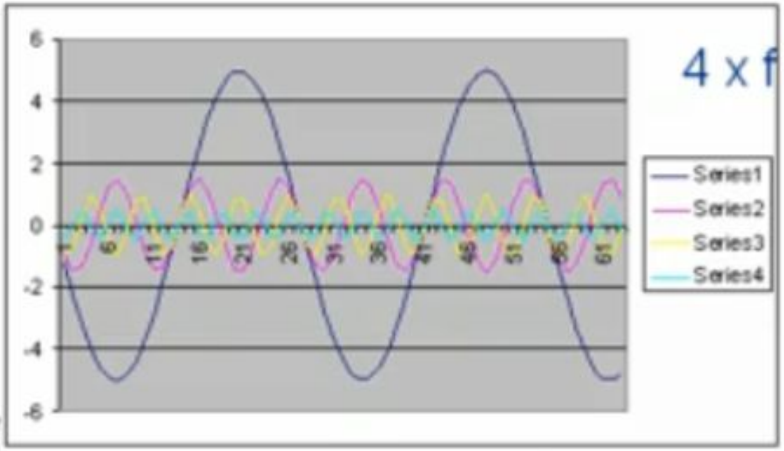


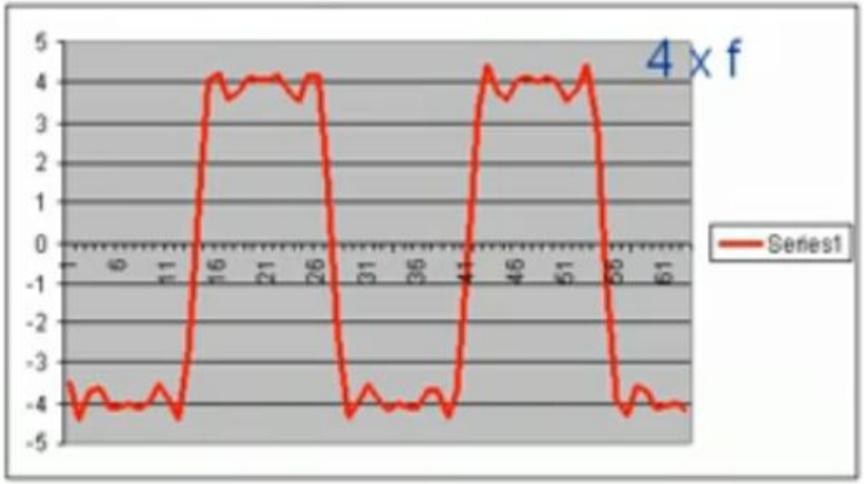




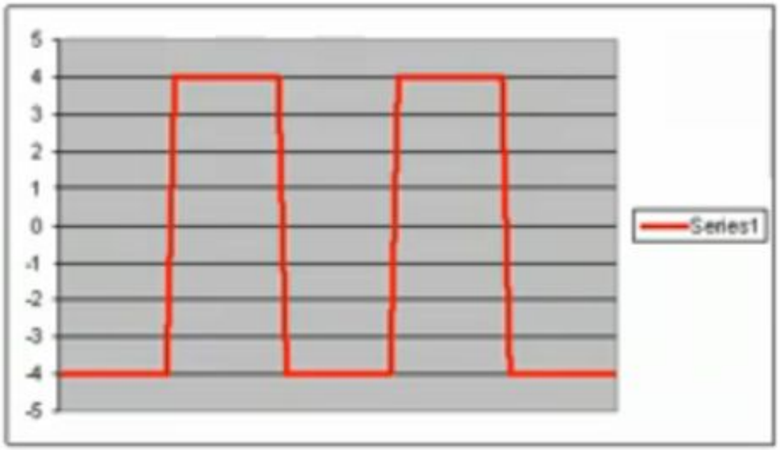
Square waves to sine waves

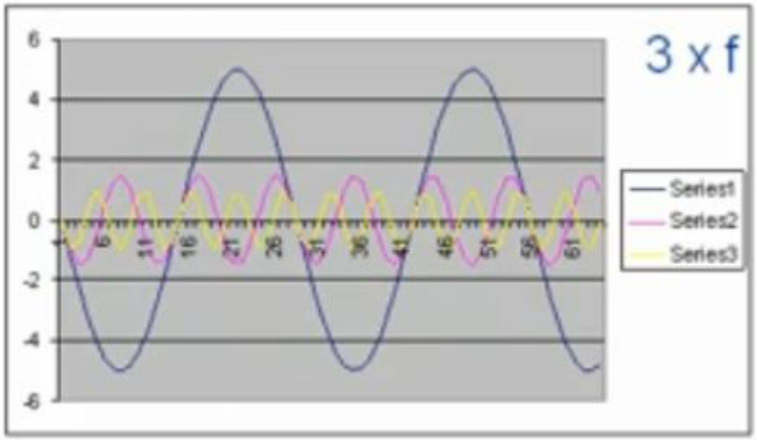


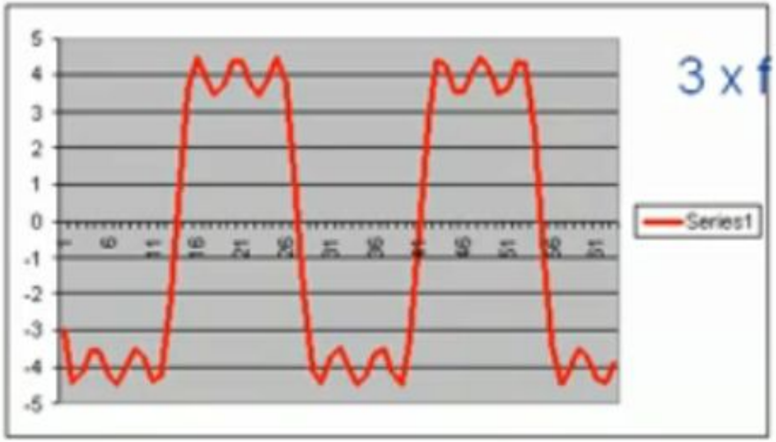




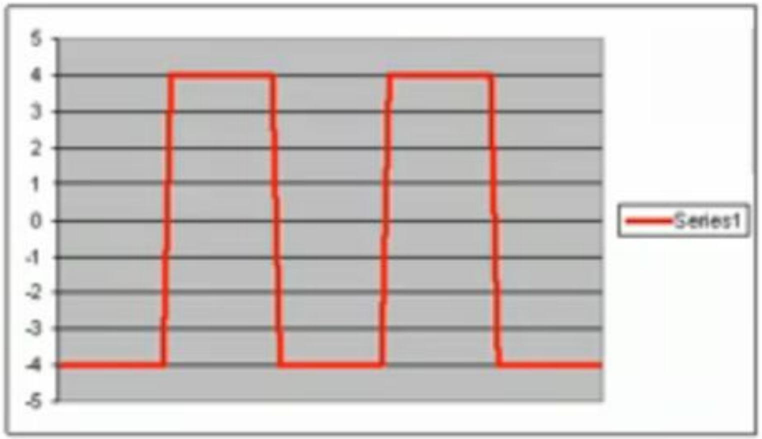
Square waves to sine waves

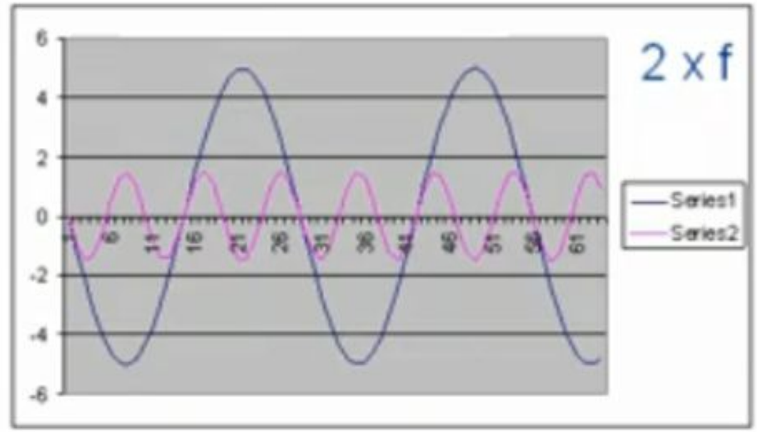


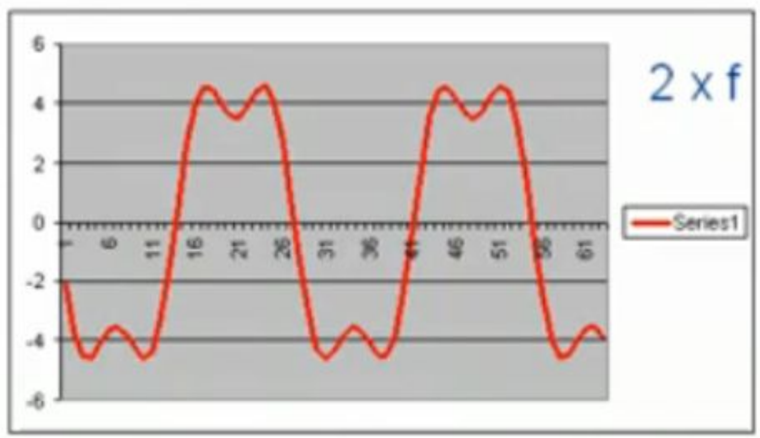




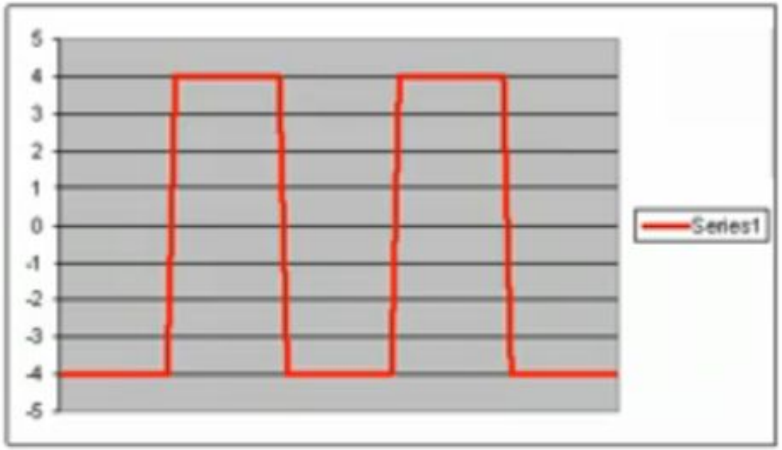
Square waves to sine waves

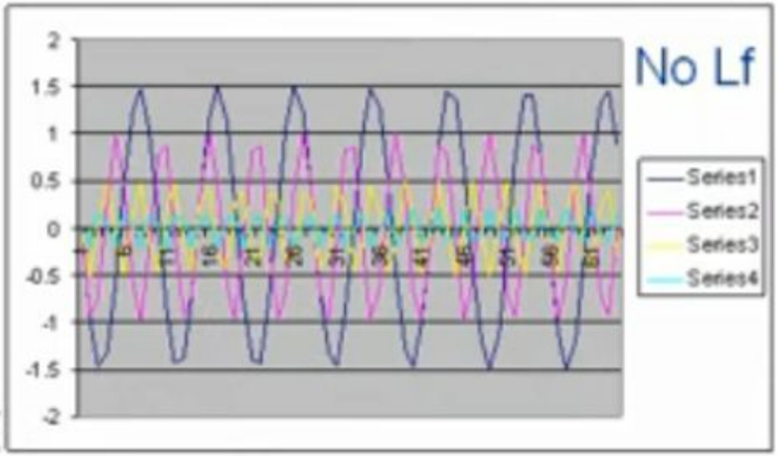


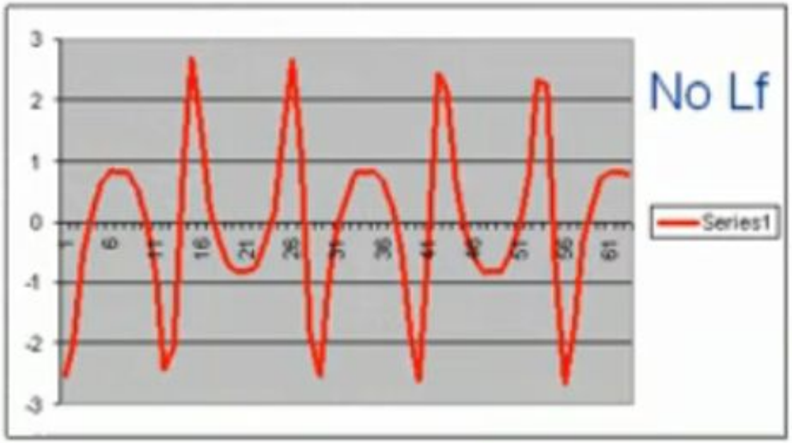




Square waves to sine waves

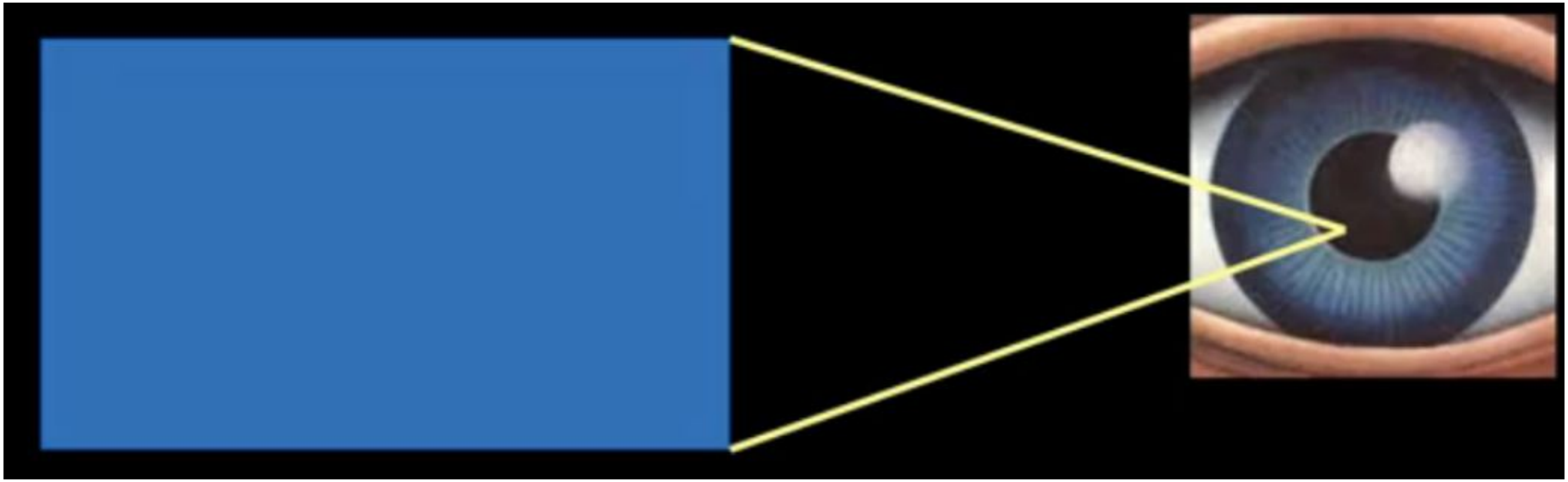






Psycho visual video compression

- There are billions of colors.
- Humans can only perceive approx 1024 shades.



JPEG Compression: Basics

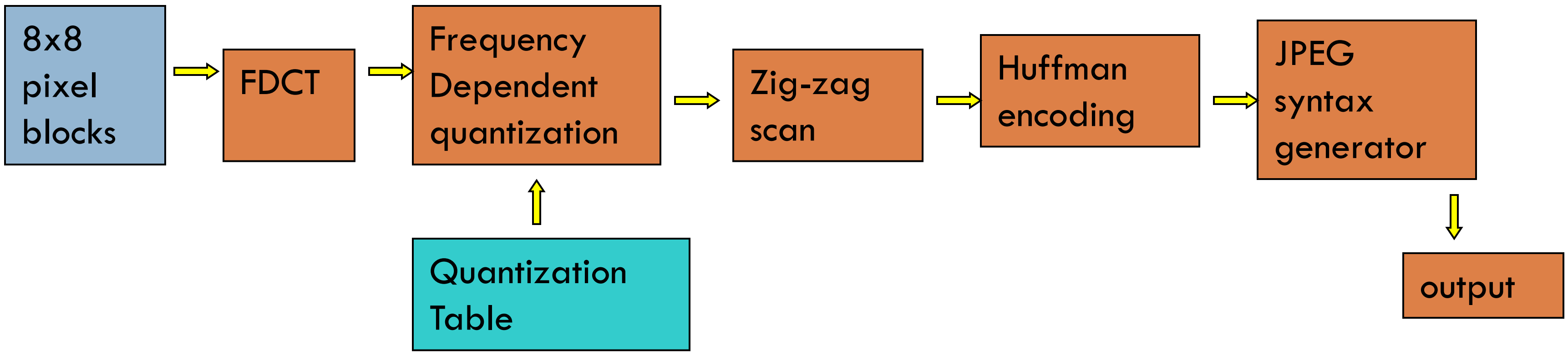
- Human vision is insensitive to high spatial frequencies
- JPEG Takes advantage of this by compressing high frequencies more coarsely and storing image as frequency data
- JPEG is a “lossy” compression scheme.

Losslessly compressed image, ~150KB

JPEG compressed, ~14KB

Flow Chart of JPEG Compression Process

- Divide image into 8x8 pixel blocks
- Apply 2D Fourier Discrete Cosine Transform (FDCT) Transform
- Apply coarse quantization to high spatial frequency components
- Compress resulting data losslessly and store



i.e. of varying JPEG compression ratios

500KB image, minimum compression

40KB image, half compression

11KB image, max compression

Close-up details of different JPEG compression ratios

Uncompressed image (roughness between pixels still visible)

Half compression, blurring & halos around sharp edges

Max compression, 8-pixel blocks apparent, large distortion in high-frequency areas

GIF 89a Image Compression

- Compuserve's image compression format
- Best for images with sharp edges, low bits per channel, computer graphics where JPEG spatial averaging is inadequate
- Usually used with 8-bit images, allowing a single image to reference a palette of up to 256 distinct colors. The colors are chosen from the 24-bit RGB color space..
- GIF images are compressed using the Lempel-Ziv-Welch (LZW)

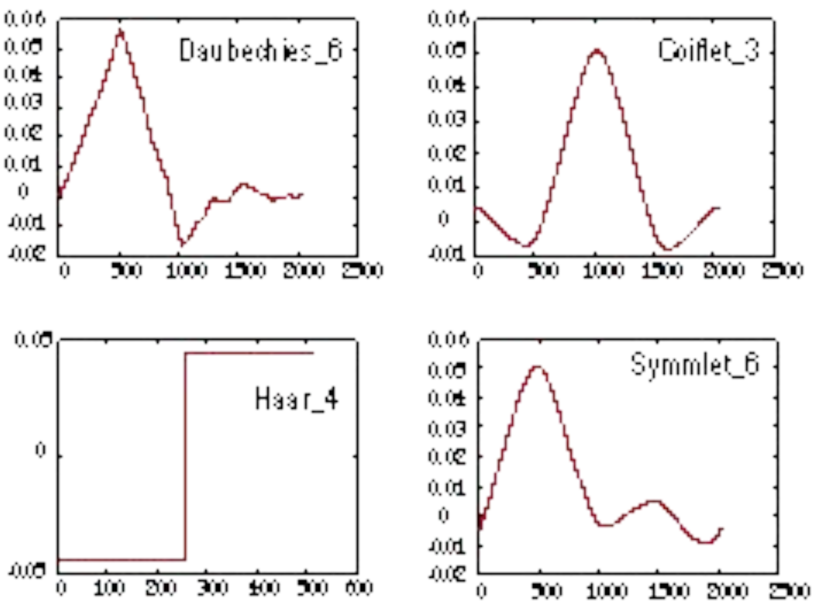
GIF 89a examples vs. JPEG

GIF Image, 7.5KB,
optimal encoding

JPEG, blotchy spots in
single-color areas

Wavelet Image Compression

- Optimal for images containing sharp edges, or continuous curves/lines (fingerprints)
- Compared with DCT, uses more optimal set of functions to represent sharp edges than cosines.
- Wavelets are finite in extent as opposed to sinusoidal functions



Several different families of wavelets.

Source: "An Introduction to Wavelets". http://www.amara.com/IEEEwave/IEEEwavelet.html#contents

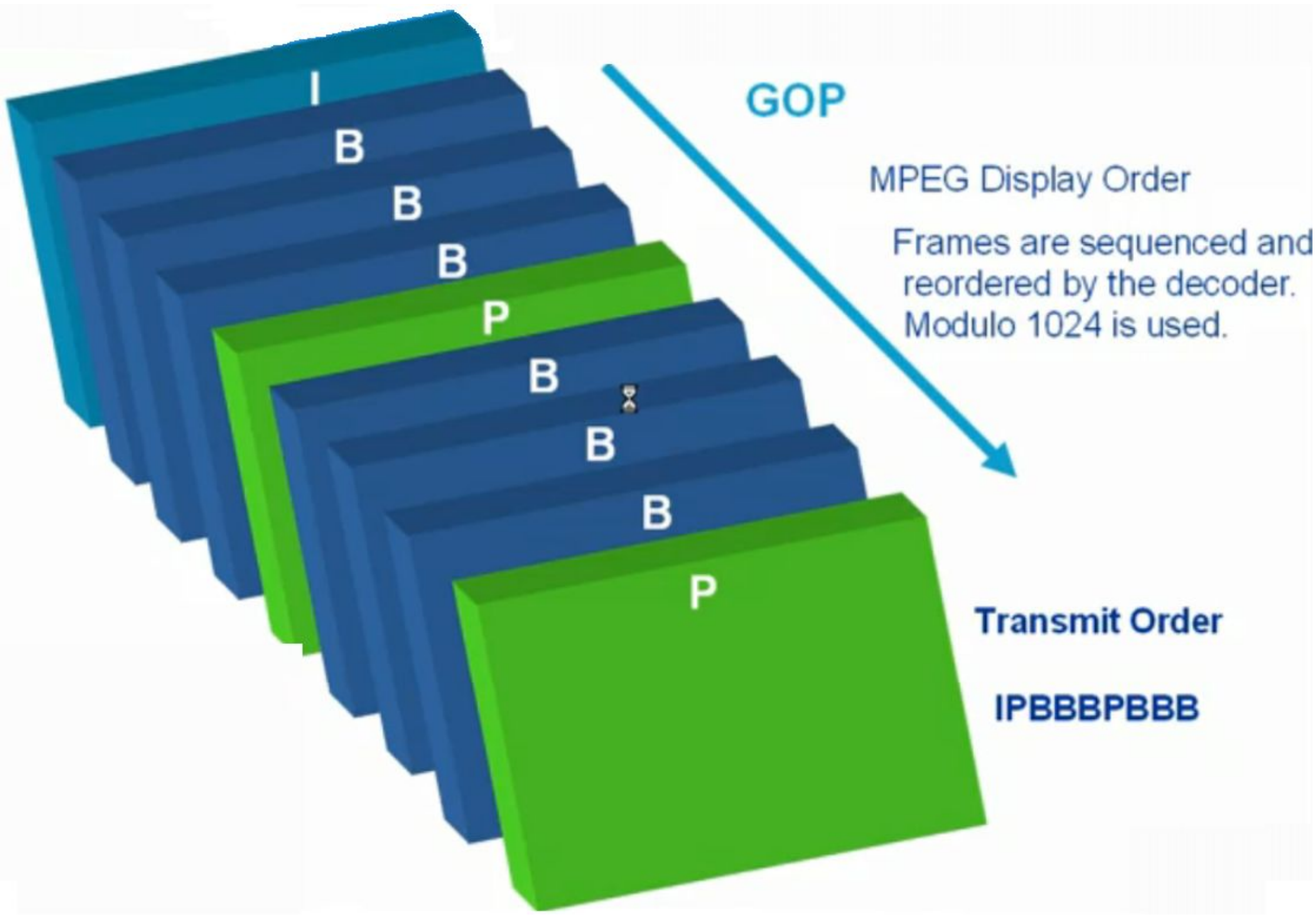
Wavelet vs. JPEG compression

Wavelet compression
file size: 1861 bytes
compression ratio - 105.6

JPEG compression
file size: 1895 bytes
compression ratio - 103.8

Source: "About Wavelet Compression". http://www.barrt.ru/parshukov/about.htm.

MPEG Frames types Group of pictures



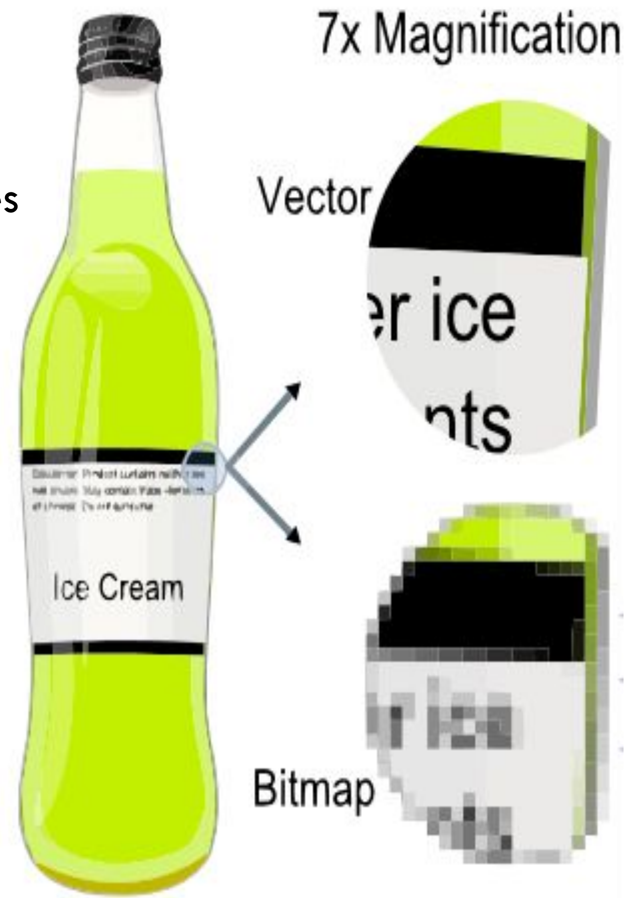
Multimedia

Images vs. Graphics

- Pictures can be encoded in a wide variety of ways
- Images are bitmaps of pixels
  - it takes scanning/rendering to produce images
  - images have a certain native resolution
  - scanning is done by a scanner, a fax, or a camera's CCD
- Vector Graphics are composed out of graphic primitives
  - graphics can be searchable, stylable, and scalable
  - the format can have different capabilities (2D vs. 3D)
- Graphics preserve model-level information
  - this only makes sense if there is a model
  - rendering and styling can be an expensive process

(e.g., video games)

  - images can be a snapshot of some specific "view" of graphics



Multimedia

- Images on the Web
- Graphics on the Web
- Video and Audio on the Web

Graphic Interchange Format (GIF)

- RFC 2046 registers the oldest graphics format on the Web
- GIF was subject of a long patent debate
  - the compression technique of GIF (LZW) had been patented by Unisys (1983)
  - Unisys wanted to get licensing fees from all commercial online uses of GIF
  - Portable Network Graphics (PNG) was developed as an effort to develop a copyright-free format
  - in 1999, Unisys changed its tactics and wanted to collect one-time fees ($5000-$7500) from all users
  - all GIF-related LZW expired in 2003/2004, so GIF is freely available now
- GIF's poor features make PNG the better choice anyway
  - 8 bit color (requires dithering for photographs), binary transparency
  - GIF's animation feature is the only thing that is not available in PNG (yet)
    - there is animated PNG support in recent versions of Firefox

Joint Photographic Experts Group (JPEG)

- RFC 2046 standardizes the second popular image format for the Web
- JPEG has been specifically designed for photographs
- it always is lossy (it cannot preserve the complete information from a random bitmap)
- it uses perception-based compression (for example, color precision is sacrificed for brightness)

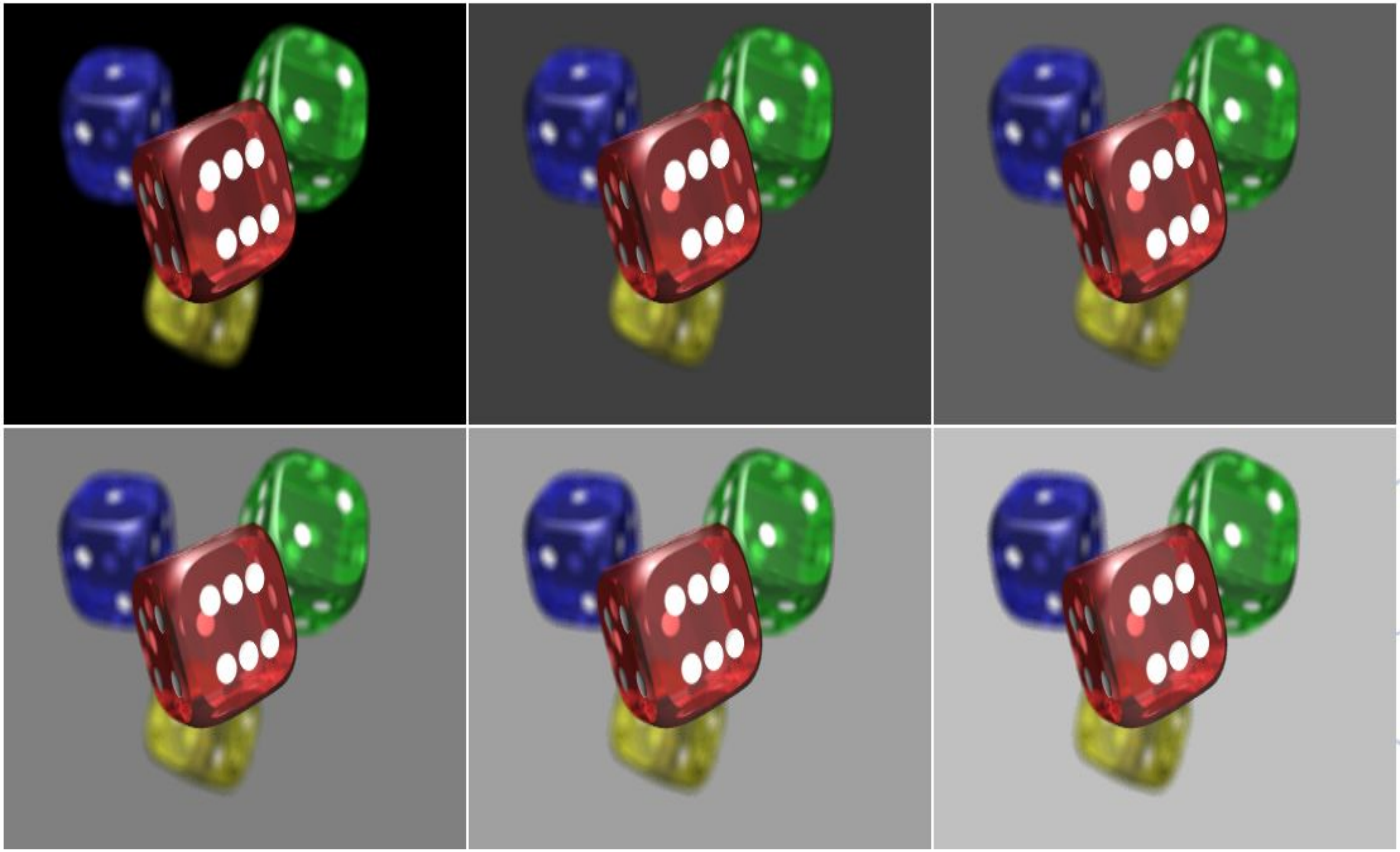
Q = 50, filesize 15,138 bytes

Q = 10, filesize 4,787 bytes

Q = 1, filesize 1,523 bytes

Portable Network Graphics (PNG)

- PNG is registered as image/png and is the third major image format
- PNG was intended to be a royalty- and copyright-free replacement of GIF
  - image formats need to supported by browsers and thus take a long time until they are established
  - IE6 implements PNG in a very rudimentary form, IE7 handles PNG correctly
- PNG has some advantages over GIF and JPEG
  - lossless, compressed palette, grayscale, or true color images
  - 8 bit alpha channel for gradual opacity (blending into the background)
- JPEG still is the preferred format for photographic pictures
- GIF still is the preferred format for animated images



JPEG, GIF and PNG

- To recap:
- JPEG is the format of choice for photographic images
- GIF was the original format of choice for drawn images such as logos or diagrams
- PNG is the new format of choice, though there are still more GIF images on the Web
  - PNG supports alpha transparency
  - GIF supports animation, which accounts for its enduring popularity
  - recent versions of Firefox have support for animated PNGs, which are superior

http://www-archive.mozilla.org/start/1.0/demos/eagle-sun.html

http://people.ischool.berkeley.edu/~ryanshaw/gucci/

http://animatedpng.com/index.php/samples/samples-by-codadude/

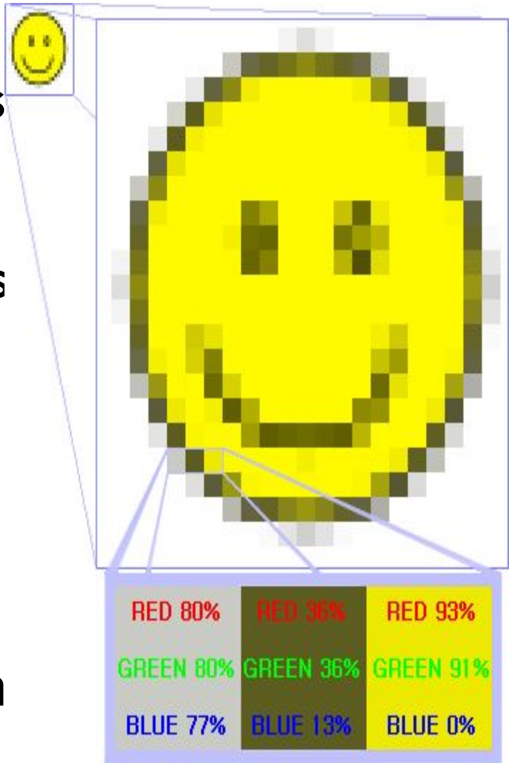
JPEG 2000

- There are no new Web image formats on the forseeable horizon
- One contender was the next generation of JPEG: JPEG 2000
- JPEG 2000 allows for progressive transmission
  - when viewing very high-resolution images, more detail can be downloaded and added to the image as the user zooms in
  - something similar is offered by Microsoft's (proprietary) Seadragon technology
- But support for JPEG 2000 on the Web seems to have stalled

http://www.seadragon.com/showcase/chris-jordan/#shipping-containers

Scalable Vector Graphics (SVG)

- JPEG, GIF and PNG are raster formats
  - they represent pictures using a grid of pixels
- SVG is a vector format
  - it represents pictures as combinations of lines and shapes
  - the lines and shapes are defined using XML
  - cannot be used for photographic images
- SVG has been a W3C recommendation since 2003 but only recently started gettin widespread browser support



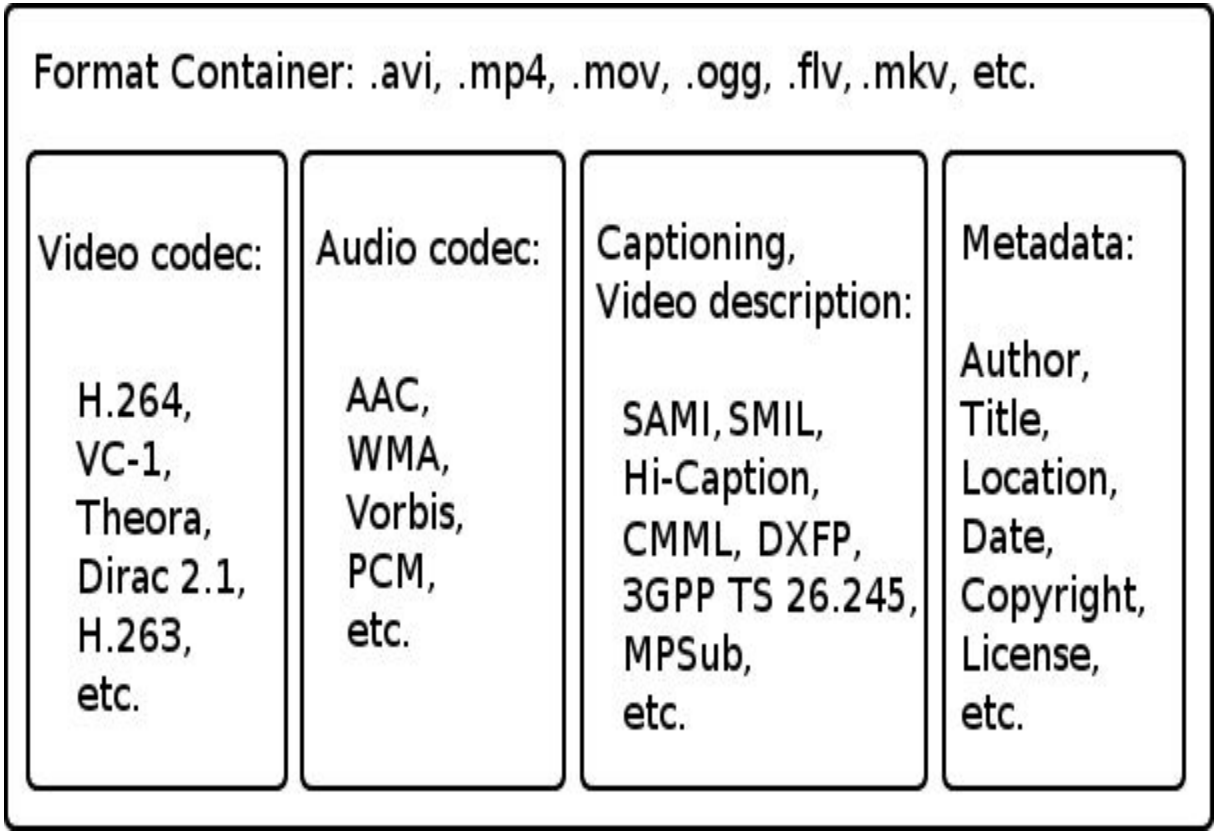
SVG Demos

- http://www.w3.org/1999/09/SVG-access/tiger.png
- http://codinginparadise.org/projects/svgweb/samples/demo.html?name=tiger&svg.render.forceflash=false
- http://vis.uell.net/gsvg/poppyr.html
- http://vis.uell.net/gsvg/electionAtlasGermany.html
- http://svg-wow.org/audio/animated-lyrics.svg

Codecs and Containers

- A codec is an algorithm used to compress a digital audio or video signal
- A container format is a particular file format used to deliver compressed audio or video
  - An audio container usually just contains one compressed audio track, plus metadata
  - A video container may contain multiple video and audio tracks, captions or subtitles, plus metadata
- Note that we don't make a distinction between codec and container for digital images
  - for example a PNG file can only hold PNG-compressed image data

Codecs and Containers



Download vs. Streaming

- Web resources usually are downloaded
  - browsers may choose to implement incremental rendering (e.g., HTML or images)
  - the resource is completely downloaded and stored
- Streaming means that there is no complete download
  - TV and phone calls are classical examples of streaming
  - some data sources cannot be downloaded (e.g., a live web cam)
- Streaming often is also used because of security issues
  - downloads make it easy to get content and redistribute it
  - streaming makes redistribution much harder (content must be destreamed)
  - the data formats for streaming are often undisclosed

Limitations in SMTP

- Only uses NVT 7 bit ASCII format
  - How to represent other data types?
- No authentication mechanisms
- Messages are sent un-encrypted
- Susceptible to misuse (Spamming, faking sender address)

base64 encoding example

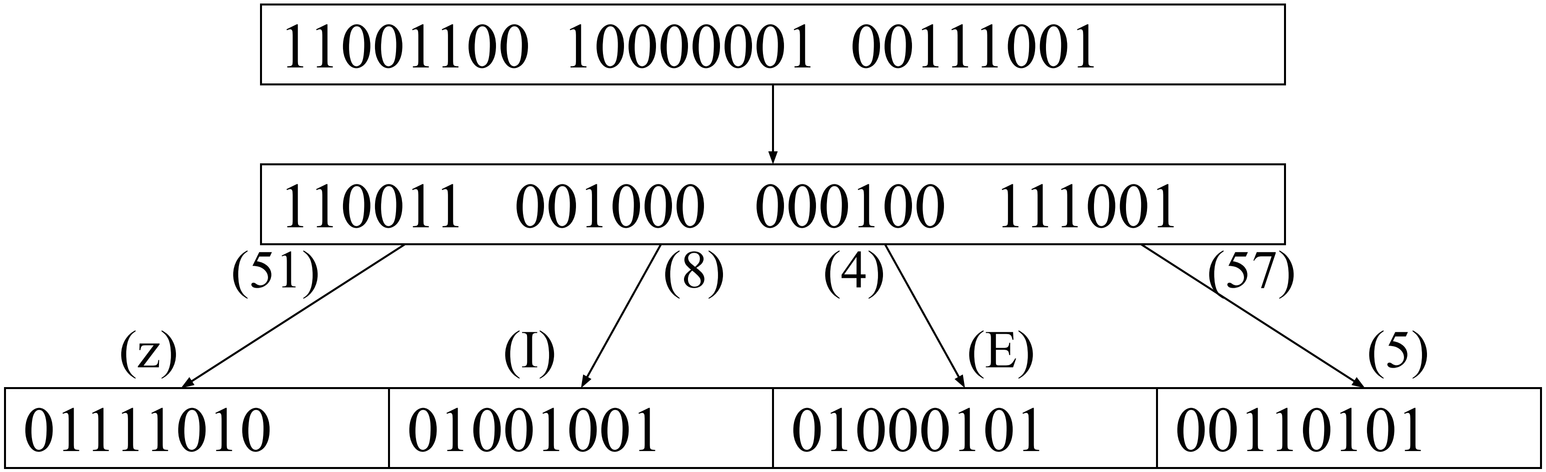
Man is distinguished, not only by his reason, but by this singular passion from other animals, which is a lust of the mind, that by a perseverance of delight in the continued and indefatigable generation of knowledge, exceeds the short vehemence of any carnal pleasure.

base64 encoded:

TWFuIGlzIGRpc3Rpbmd1aXNoZWQsIG5vdCBvbmx5IGJ5IGhpcyByZWFzb24sIGJ1dCBieSB0
aGlzIHNpbmd1bGFyIHBhc3Npb24gZnJvbSBvdGhlciBhbmltYWxzLCB3aGljaCBpcyBhIGx1
c3Qgb2YgdGhlIG1pbmQsIHRoYXQgYnkgYSBwZXJzZXZlcmFuY2Ugb2YgZGVsaWdodCBpbiB0
aGUgY29udGludWVkIGFuZCBpbmRlZmF0aWdhYmxlIGdlbmVyYXRpb24gb2Yga25vd2xlZGdl
LCBleGNlZWRzIHRoZSBzaG9ydCB2ZWhlbWVuY2Ugb2YgYW55IGNhcm5hbCBwbGVhc3VyZS4=

Base64 Encoding

- Divides binary data into 24 bit blocks
- Each block is then divided into 6 bit chunks
- Each 6-bit section is interpreted as one character, 25% overhead



Encoding of 6-bit values (base-64 encoding).

6-bit value	ASCII char	6-bit value	ASCII char	6-bit value	ASCII char	6-bit value	ASCII char
0	A	10	Q	20	g	30	w
1	B	11	R	21	h	31	x
2	C	12	S	22	i	32	y
3	D	13	T	23	j	33	z
4	E	14	U	24	k	34	0
5	F	15	V	25	l	35	1
6	G	16	W	26	m	36	2
7	H	17	X	27	n	37	3
8	I	18	Y	28	o	38	4
9	J	19	Z	29	p	39	5
a	K	1a	a	2a	q	3a	6
b	L	1b	b	2b	r	3b	7
c	M	1c	c	2c	s	3c	8
d	N	1d	d	2d	t	3d	9
e	O	1e	e	2e	u	3e	+
f	P	1f	f	2f	v	3f	/